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1. Protostele:

In protostele phloem surrounds xylem. The type includes Haplostele, Actinostele, Plectostele, and Mixed protostele.

(i) **Haplostele**: Xylem surrounded by phloem is known as haplostele. Example: *Selaginella.*

(ii) **Actinostele**: Star shaped xylem core is surrounded by phloem is known as actinostele. Example: *Lycopodium serratum.*

(iii) **Plectostele**: Xylem plates alternates with phloem plates. Example: *Lycopodium clavatum.*

(iv) **Mixed prototostele**: Xylem groups uniformly scattered in the phloem. Example: *Lycopodium cernuum.*

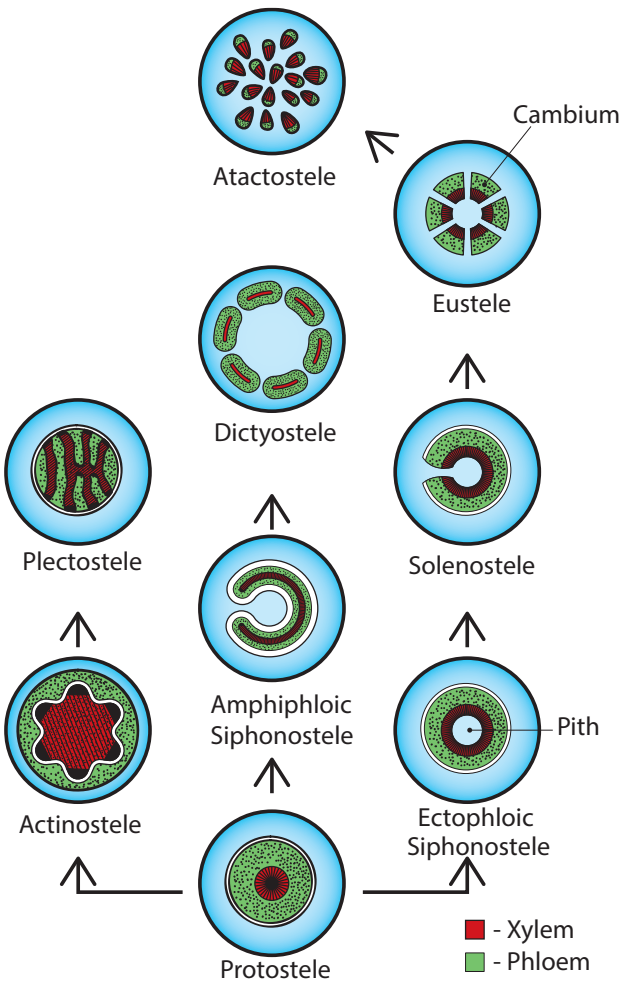


Figure 2.37: Types of Stele

2. Siphonostele:

In siphonostele xylem is surrounded by phloem with pith at the centre. It includes Ectophloic siphonostele, Amphiphloic siphonostele, Solenostele, Eustele, Atactostele and Polycylic stele.

(i) Ectophloic siphonostele: The phloem is restricted only on the external side of the xylem. Pith is in centre. Example: *Osmunda.*

(ii) Amphiphloic siphonostele: The phloem is present on both the sides of xylem. The pith is in the centre. Example: *Marsilea.*

(iii) Solenostele: The stele is perforated at a place or places corresponding the origin of the leaf trace.

(a) Ectophloic solenostele – Pith is in the centre and the xylem is surrounded by phloem Example *Osmunda.*

(b) Amphiphloic solenostele – Pith is in the centre and the phloem is present on both sides of the xylem. Example: *Adiantum pedatum.*

(c) Dictyostele – The stele is separated into several vascular strands and each one is called meristele. Example: *Adiantum capillus-veneris.*

(iv) Eustele: The stele is split into distinct collateral vascular bundles around the pith. Example: Dicot stem.

(v) Atactostele: The stele is split into distinct collateral vascular bundles and are scattered in the ground tissue Example: Monocot stem.

(vi) Polycyclicstele: The vascular tissues are present in the form of two or more concentric cylinders. Example: *Pteridium.*